

Lewis acids and bases (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to:

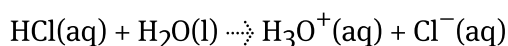
- Apply Lewis acid–base theory to inorganic and organic chemistry to identify the role of the reacting species.
- Explain the relationship between Brønsted–Lowry acids and bases and Lewis acids and bases.

Section R3.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/>) covers the Brønsted–Lowry acid and base interaction in which a proton must be transferred. This theory works well for common acids and bases in which molecules accept and donate protons. However, this theory does not hold for all acid–base interactions. One example is the identification of iron(III) ions as acidic in water. How can these ions show acidity if they do not have any protons to donate? What about the participation of electron pairs in determining the acidic and basic behaviour of molecules? Electron pairs play an important role because when a molecule donates or accepts protons, it leaves behind or gains an electron pair.

These exceptions can be explained by the Lewis acids and bases theory, which focuses on electron transfer rather than proton transfer. This concept is more general, so it can cover most substances. The interaction of electron pairs plays a significant role in identifying both a Lewis acid and Lewis base during a reaction. What is the role of a lone pair in Lewis acids and bases? How do you identify them? Is there any relationship between Brønsted–Lowry acids and bases and Lewis acids and bases?

Definition of a Lewis acid and base

Before discussing the definition of Lewis acids and bases, we will review the concept of Brønsted–Lowry acids and bases, in which a proton donor substance is called an acid and a proton acceptor substance is called a base. For example, the reaction between hydrochloric acid and water.



This reaction shows that a proton is donated by the hydrochloric acid as an acid to water, which acts as the base forming a hydronium ion.

Now, think about how this proton transfers from the perspective of the electron movement for the breaking and forming of covalent bonds (**Figure 1**). This process also involves transfer of an electron as the partial negative charge oxygen atom in water attracts the partial positively charged hydrogen atom in hydrogen chloride, causing the covalent bond in this molecule to break. As a result, the hydrogen atom as a proton forms a new bond with the oxygen atom, forming a hydronium ion. The chlorine atom is then left with an extra electron that becomes a chloride ion.

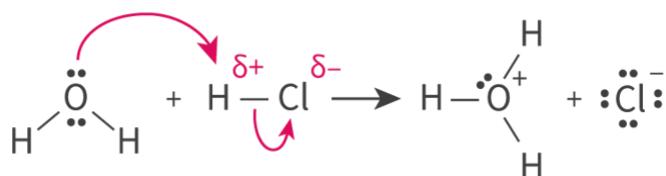


Figure 1. Proton transfer mechanism.

 More information for figure 1

The diagram illustrates the mechanism of proton transfer in a chemical reaction between water (H_2O) and hydrogen chloride (HCl). On the left, a water molecule (H_2O) is shown with two hydrogen atoms and one oxygen atom, which has two lone pairs of electrons. The oxygen has an overall neutral charge.

In the middle, an HCl molecule is depicted, with a partial positive charge (δ^+) on the hydrogen and a partial negative charge (δ^-) on the chlorine, indicating a polar covalent bond.

The diagram uses curved arrows to indicate the movement of electrons. A curved arrow goes from one of the lone pairs on the oxygen in water to the hydrogen in HCl , showing that the oxygen donates an electron pair to form a bond with the hydrogen, creating a hydronium ion (H_3O^+) on the right.

A parallel arrow shows the breaking of the bond between H and Cl in HCl , transferring an electron to the chlorine atom, resulting in a chloride ion (Cl^-), which is depicted with an extra pair of electrons.

[Generated by AI]

This process shows how electron transfer is crucial in chemical reactions, which leads to the Lewis acid–base theory based on electron-pair transfer. The interaction of electrons from reactant molecules leads to formation of a new chemical as a pair of electrons is donated by Lewis base and received by the Lewis acid (**Figure 2**). This theory defines acid as a substance that can accept a lone pair and base is a substance that can donate a lone pair.

Common examples of Lewis acids are protons (H^+ ion), cations of metals such as magnesium ion, cations of d-block elements such as a silver ion, atoms that have incomplete valence shells (such as aluminium and boron) and carbocation and

carbonyl compounds.



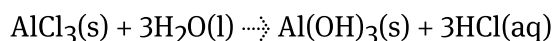
Figure 2. Formation of coordination bond by the interaction between a Lewis acid (A) and a Lewis base (B).

 More information for figure 2

The diagram illustrates the formation of a coordination bond through the interaction of a Lewis acid and a Lewis base. On the left, the letter 'A' is labeled as a Lewis acid, and to its right, ':B' is labeled as a Lewis base. An arrow extends from 'A' to ':B', indicating the donation of an electron pair by the Lewis base to the Lewis acid. The result of this interaction is shown to the right with 'A - B', representing the newly formed coordination bond.

[Generated by AI]

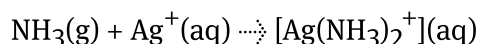
For example, AlCl_3 has an incomplete valence shell although it shares its three electrons with chlorine atoms and forms three single bonds. Therefore, it is still able to receive electron pairs and this is a behaviour of Lewis acid.



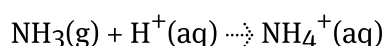
On the other hand, the Lewis base is commonly characterised as an electron-rich or negatively charged atom. The electron pairs that are accepted by the Lewis acids from Lewis bases are used to form a covalent bond. Common examples of Lewis bases are water, ammonia, hydroxide ions, cyanide ions, halide ions and ethanoate ions.

Lewis acid—base theory with inorganic and organic examples

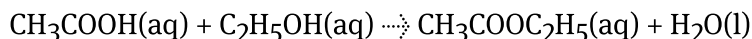
Looking at the previous examples, this Lewis acid–base theory can be applied to both organic and inorganic chemistry. In inorganic chemistry, metal cations commonly act as Lewis acids due to their positive charge. For example, ammonia acts as a Lewis base and donates a pair of electrons to the silver ion or hydrogen ion as a Lewis base.



or



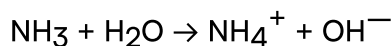
Whereas, in organic chemistry, carbocation or carbonyl compounds often act as Lewis acids as they have an electron-deficient carbon atom that accepts an electron pair from nucleophiles such as hydroxide ions or alcohol. This occurs in esterification reactions where an ester is formed when an alcohol is reacted with a carboxylic acid.



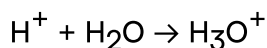
Study skills

Water can be both identified as a Lewis acid or a Lewis base, depending on the reaction it is involved in. Therefore, water is also said to be amphoteric (see [subtopic R3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/)).

It acts as a Lewis acid and receives a pair of electrons from ammonia.



It acts as a Lewis base and donates a pair of electrons to hydrogen ions.



The relationship between Brønsted—Lowry and Lewis acid—base theory

In comparison with the Brønsted—Lowry acids and bases (see [section R3.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/)), Lewis acids and bases theory is broader as it is not restricted to just those reactions that include the transfer of protons or water as solvents.

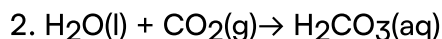
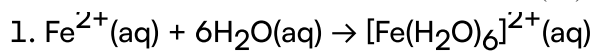
In fact, all Brønsted—Lowry acids are also considered Lewis acids because of the availability to donate protons and therefore are also able to accept an electron pair. On the other hand, not all Lewis acids are Brønsted—Lowry acids. For example, a transition metal ion in a complex ion can act as a Lewis acid by accepting an electron pair, but it cannot act as a Brønsted—Lowry acid, as it cannot donate a proton.

Similarly, all Brønsted—Lowry bases are considered Lewis bases as all Brønsted—Lowry bases have a lone pair to accept hydrogen and all Lewis bases have lone pairs. Conversely, all Lewis bases are not Brønsted—Lowry bases as Lewis bases can act as nucleophiles, rather than Brønsted—Lowry bases.

Try **Worked example 1** and the exercise question on the Lewis acid—base theory.

Worked example 1

Identify and explain the role of each reactant in these reactions according to the Lewis acid—base theory.



Lewis acids: _____

Lewis bases: _____

Lewis acids: 1. Fe^{2+} and 2. CO_2

Lewis bases: 1. and 2. H_2O

The iron(II) ion and carbon dioxide act as a Lewis acid as they accept an electron pair from water as a Lewis base, forming a new bond.

Activity

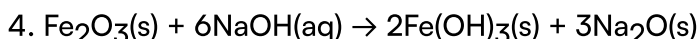
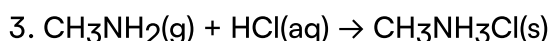
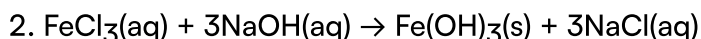
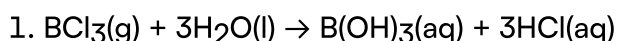
- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
 - Social skills — Working collaboratively to achieve a common goal
 - Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

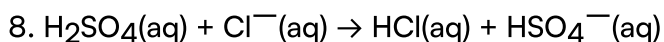
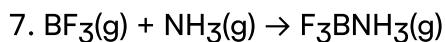
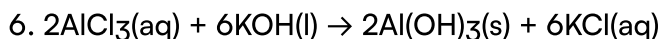
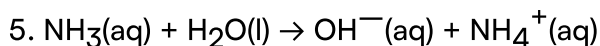
Your task

With your partner, identify the reactants in the reactions below and state whether each one can be categorised as a Lewis acid, Lewis base, Brønsted—Lowry acid or Brønsted—Lowry base. Then identify Lewis acids that are not Brønsted—Lowry acids and Lewis bases that are not Brønsted—Lowry bases from the same reactions, and complete the table provided in the worksheet.

Worksheet 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry%20Worksheet%20Lewis%20Acids%20and%20Bases.pdf)





Discuss and draw a conclusion with your partner as to what the similarities and differences are between Lewis acid—base theory and Brønsted—Lowry acid—base theory. Also, discuss the differences and similarities between Lewis acids and Brønsted—Lowry acids, and between Lewis bases and Brønsted—Lowry bases.

Brønsted—Lowry acids and Lewis acids

In reaction 1, BCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 2, FeCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 3, HCl is a Lewis acid and a Brønsted—Lowry acid.

In reaction 4, Fe_2O_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 5, H_2O is a Lewis acid and a Brønsted—Lowry acid.

In reaction 6, AlCl_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 7, BF_3 is a Lewis acid but not a Brønsted—Lowry acid because it is not acting as a proton donor.

In reaction 8, H_2SO_4 is a Lewis acid and a Brønsted—Lowry acid.

Brønsted—Lowry bases and Lewis bases

In reaction 1, H_2O is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 2, NaOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 3, CH_3NH_2 is a Lewis base and a Brønsted—Lowry base.

In reaction 4, NaOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 5, H_2O is a Lewis base and a Brønsted—Lowry base.

In reaction 6, KOH is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 7, NH_3 is a Lewis base but not a Brønsted—Lowry base because it is not acting as a proton acceptor.

In reaction 8, Cl^- is a Lewis base and a Brønsted—Lowry base.